



School:Campus:

Academic Year: Subject Name: Subject Code:

Semester: Program: Branch: Specialization:

Date:

Applied and Action Learning

(Learning by Doing and Discovery)

Name of the Experiment: Chains Beyond Ethereum – Platform Comparisons

Objective/Aim:

- To study and compare blockchain platforms beyond Ethereum, focusing on architecture, consensus mechanisms, smart contract capabilities, and scalability.
- To explore various alternative blockchain networks and their ecosystems.
- To analyze technical differences, advantages, and limitations of each platform.
- To understand real-world use cases and suitability for decentralized applications (DApps).
- To compare consensus algorithms, transaction throughput, and developer tooling/environments.

Apparatus/Software Used:

- Web browser: used for accessing blockchain documentation and related data.
- Blockchain explorers (for example, Solscan): to view public blockchain transactions and network information.
- Crypto wallets (optional) such as MetaMask, Phantom, or Keplr: to understand and simulate blockchain interactions.
- Stable internet connection: required for researching and accessing online blockchain resources.

Theory/Concept:

- Ethereum is a Layer 1 blockchain that introduced smart contracts—self-executing programs running on a decentralized network. This innovation enabled developers to build decentralized applications (dApps), marking a shift from simple digital currencies like Bitcoin to programmable blockchains.
- Despite its impact, Ethereum faces issues such as high gas fees, limited scalability, and network congestion. To address these challenges, several Layer 1 alternatives—like Solana, Avalanche, Cardano, Polkadot, and NEAR—have emerged.
- These newer platforms focus on delivering faster transactions, lower costs, and enhanced scalability while maintaining security and decentralization.
- Each blockchain offers distinct architectures and consensus mechanisms designed to overcome Ethereum's limitations and expand blockchain applications across different sectors.

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*** As applicable according to the experiment.**

Two sheets per experiment (10-20)

Observation table:

Blockchain	Consensus Mechanism	Transaction Speed (TPS)	Key Features	Primary Use Case
Bitcoin (BTC)	Proof of Work	~7	Most secure and decentralized	Digital currency and store of value
BNB Chain (BNB)	Proof of Staked Authority	~160	EVM compatible, low fees	DeFi, gaming, token trading
Solana (SOL)	Proof of History + PoS	~65,000	High throughput, low latency	NFTs, DeFi, fast dApps
Avalanche (AVAX)	Avalanche Consensus (PoS)	~4,500	Custom subnets, flexibility	Financial dApps, enterprise apps
Cardano (ADA)	Ouroboros Proof of Stake	~250	Research-based, energy-efficient	Identity, governance, education
Polkadot (DOT)	Nominated Proof of Stake	~1,000	Parachains, cross-chain support	Interoperability between blockchains
Cosmos (ATOM)	Tendermint Proof of Stake	~1,000	IBC protocol for app chains	Cross-chain communication
NEAR Protocol (NEAR)	Sharded Proof of Stake	~100,000 (theoretical)	Human-readable accounts, scalable	Web3 applications, user-friendly dApps
Algorand (ALGO)	Pure Proof of Stake	~1,000	Fast finality, low energy	Payments, finance, sustainability
Tezos (XTZ)	Liquid Proof of Stake	~40	Self-amending governance model	NFTs, art, on-chain voting

*Conclusion:

- This study highlights how Layer 1 blockchains beyond Ethereum have expanded the blockchain ecosystem by addressing Ethereum's limitations.
- Solana and NEAR emphasize speed and high performance.
- Cardano and Algorand focus on research-driven development and energy efficiency.
- Polkadot and Cosmos promote cross-chain interoperability.
- BNB Chain and Avalanche offer flexible and developer-friendly environments.
- Collectively, these platforms are shaping a multi-chain blockchain ecosystem that is more scalable, accessible, and environmentally sustainable.

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	50		

Signature of the Student :

Name :

Signature of the Faculty :

Regn. No. :

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Two sheets per experiment (10-20) to be used***

